

Capstone Project - Digital Marketing

The requirement is to predict the conversion rate (0 or 1) based on several parameters

As a data scientist, I need to develop a model to predict the conversion rate

1.) Identify your problem statement

There are three stages in model prediction

1. Machine learning
2. Supervised learning
3. Classification

2.) Basic Info about dataset

No of rows: 8000 No of columns: 20

3.) Preprocessing methods

- Converted categorical data into numerical data
- Created binary columns to avoid redundancy

4. Machine Learning models

1. SVM

The f1_macro value for best parameter {'C': 100, 'gamma': 'scale', 'kernel': 'rbf'}: **0.8715212267364234**

The confusion Matrix:

```
[[ 87 107]
 [ 96 1310]]
```

The report:

	precision	recall	f1-score	support
0	0.48	0.45	0.46	194
1	0.92	0.93	0.93	1406
accuracy			0.87	1600
macro avg	0.70	0.69	0.69	1600
weighted avg	0.87	0.87	0.87	1600

ROC_AUC_Score: **0.8604306286753384**

2. Naive Bayes

The f1_macro value for best parameter {'alpha': 10}: 0.02748725993964525

The confusion Matrix:

```
[[ 194  0]
 [1405  1]]
```

The report:

	precision	recall	f1-score	support
0	0.12	1.00	0.22	194
1	1.00	0.00	0.00	1406
accuracy			0.12	1600
macro avg	0.56	0.50	0.11	1600
weighted avg	0.89	0.12	0.03	1600

ROC_AUC_Score: 0.6767681952163775

3. Decison Tree

The f1_macro value for best parameter {'criterion': 'entropy', 'max_features': 'log2', 'splitter': 'best'}: 0.06444983652727414

The confusion Matrix:

```
[[ 193  1]
 [1375 31]]
```

The report:

	precision	recall	f1-score	support
0	0.12	0.99	0.22	194
1	0.97	0.02	0.04	1406
accuracy			0.14	1600
macro avg	0.55	0.51	0.13	1600
weighted avg	0.87	0.14	0.06	1600

ROC_AUC_Score: 0.5084468624891848

4. Logistic Regression

The f1_macro value for best parameter {'penalty': 'l1', 'solver': 'saga'}: 0.795516573295985

The confusion Matrix:

```
[[ 146  48]
 [ 340 1066]]
```

The report:

	precision	recall	f1-score	support
0	0.30	0.75	0.43	194
1	0.96	0.76	0.85	1406
accuracy			0.76	1600
macro avg	0.63	0.76	0.64	1600
weighted avg	0.88	0.76	0.80	1600

ROC_AUC_Score: 0.8415113431391238

5. KNN

The f1_macro value for best parameter {'metric': 'minkowski', 'n_neighbors': 7, 'p': 2}: 0.7683264533883728

The confusion Matrix:

```
[[ 122  72]
 [ 368 1038]]
```

The report:

	precision	recall	f1-score	support
0	0.25	0.63	0.36	194
1	0.94	0.74	0.83	1406
accuracy			0.72	1600
macro avg	0.59	0.68	0.59	1600
weighted avg	0.85	0.72	0.77	1600

ROC_AUC_Score: 0.7524013432857708

6. Random Forest

The f1_macro value for best parameter {'criterion': 'log_loss', 'max_features': 'sqrt', 'n_estimators': 10}:
0.0627935606060606

The confusion Matrix:

```
[[ 190   4]
 [1376  30]]
```

The report:

	precision	recall	f1-score	support
0	0.12	0.98	0.22	194
1	0.88	0.02	0.04	1406
accuracy			0.14	1600
macro avg	0.50	0.50	0.13	1600
weighted avg	0.79	0.14	0.06	1600

ROC_AUC_Score: 0.49459056180434363

Based on the above models, SVM is the best to evaluate the conversion rate as the accuracy is
0.8715212267364234 & ROC value is **0.8604306286753384**